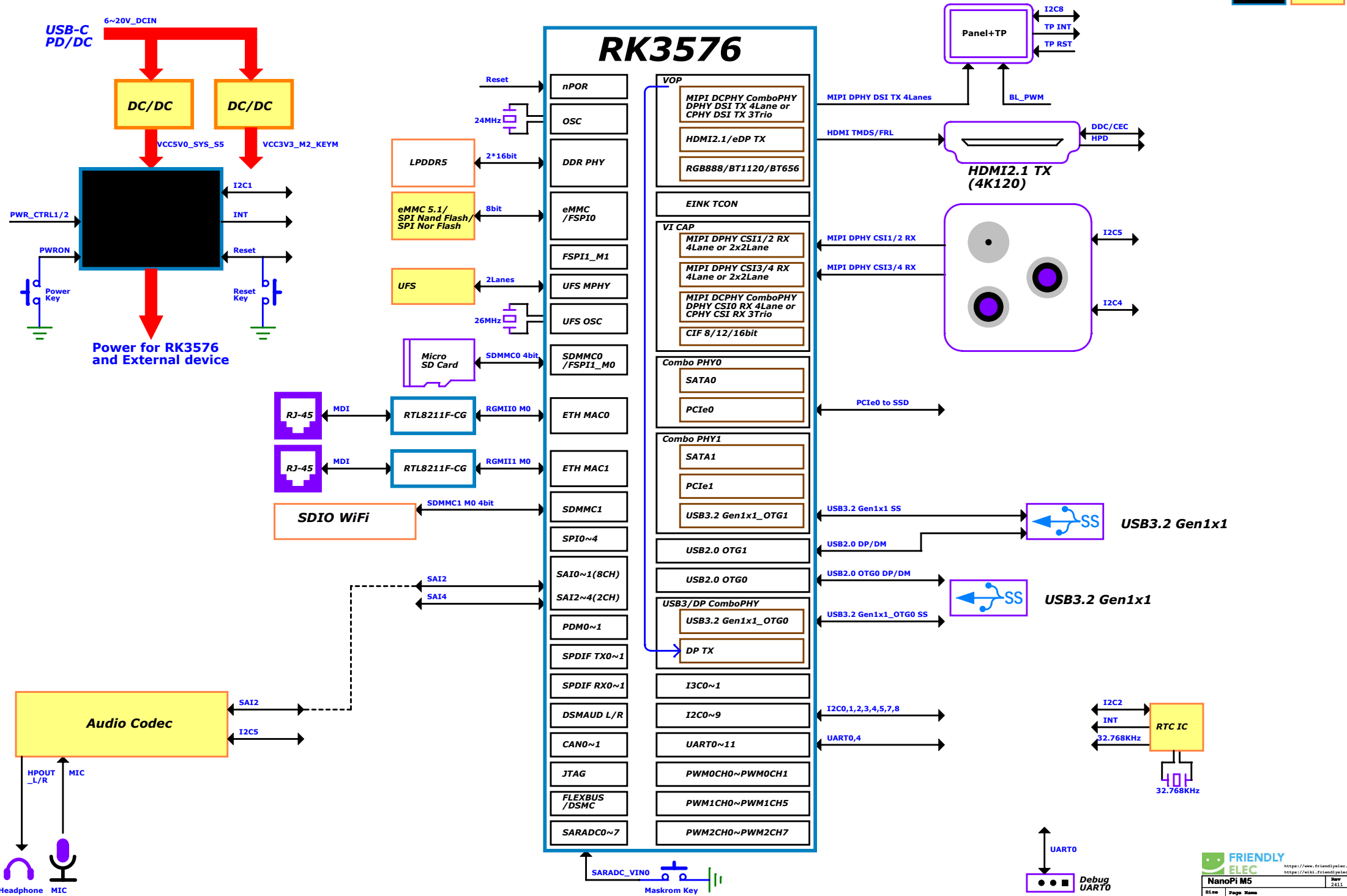
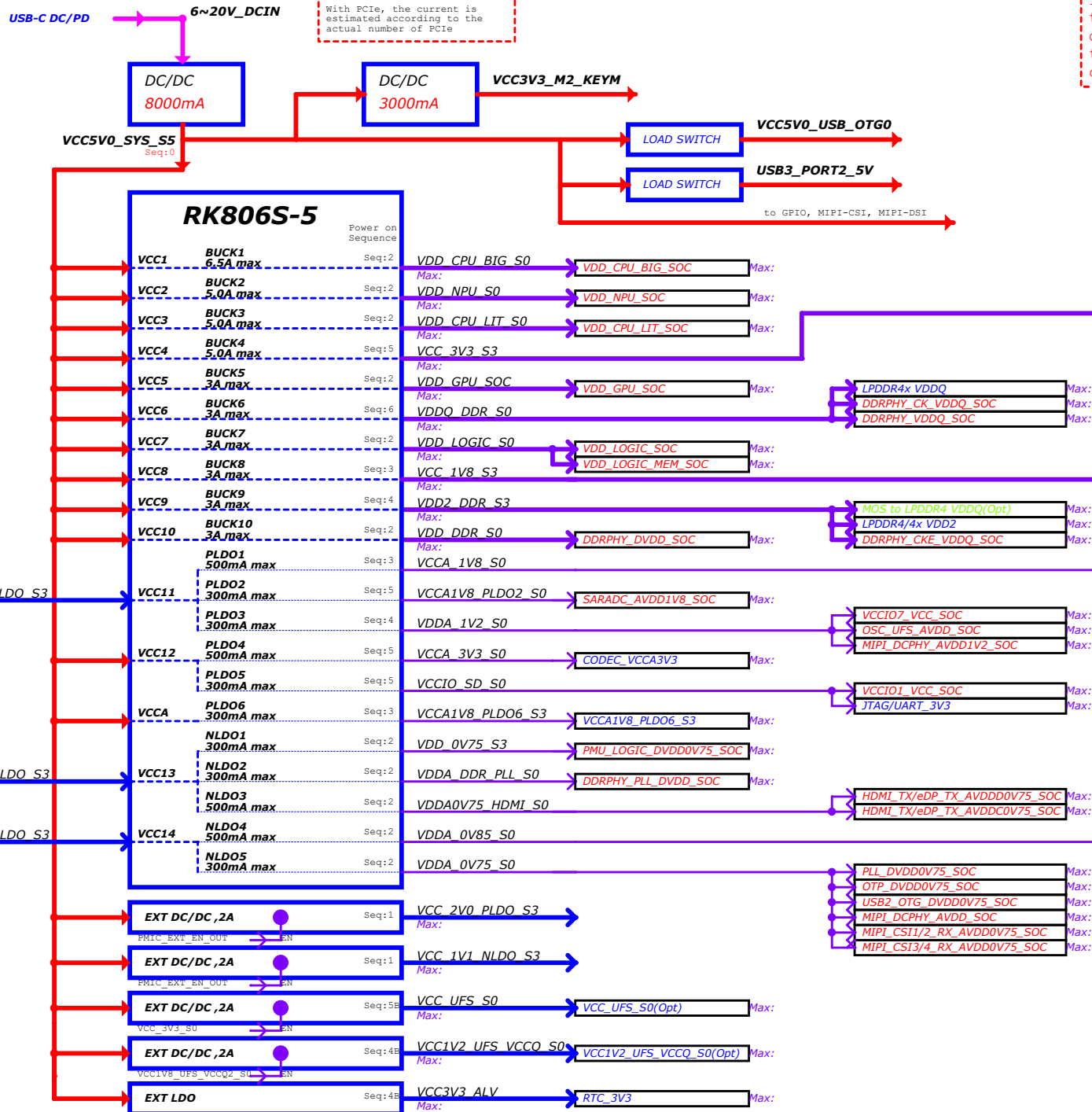


NanoPi M5

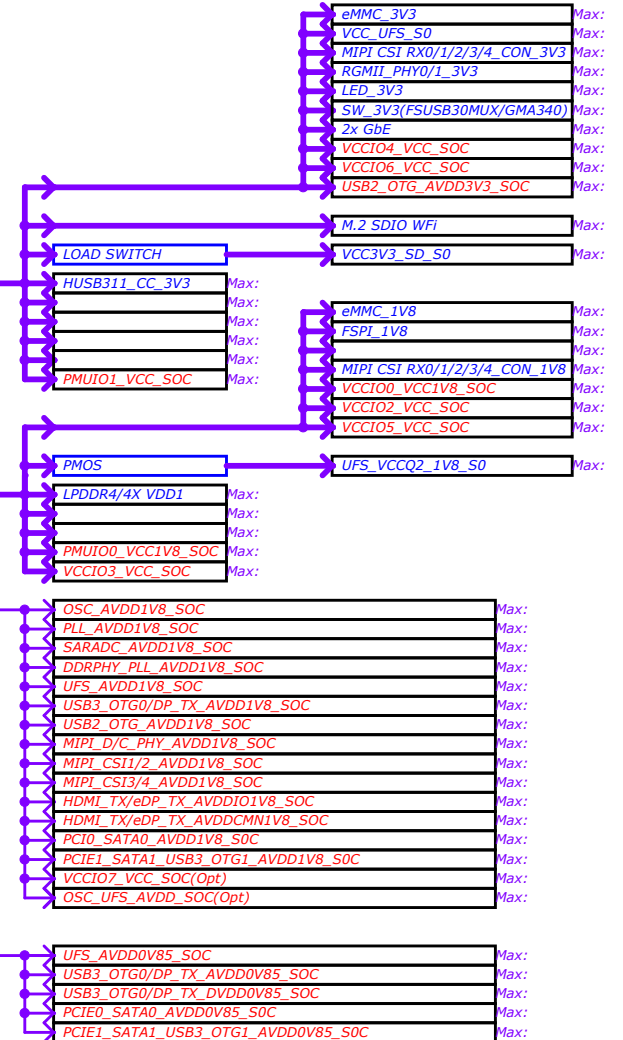


Power Tree

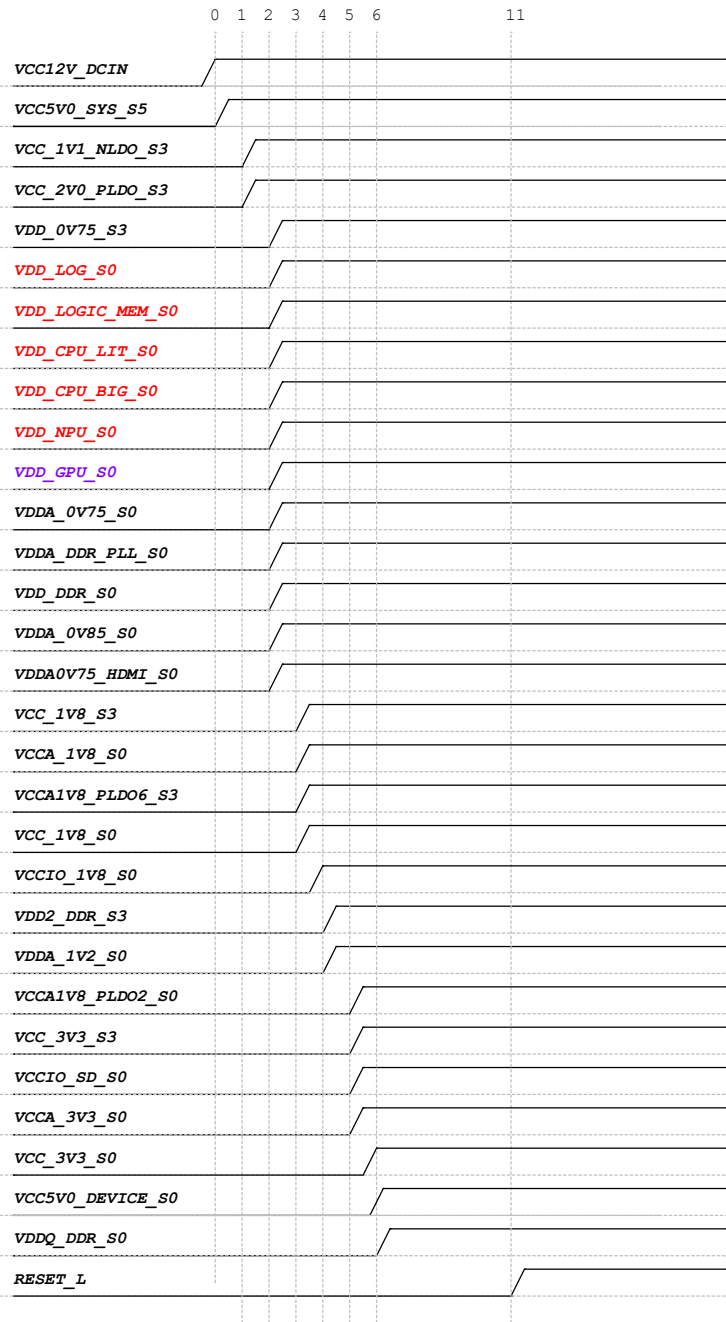


Note:

The RK806S-J LDO power distribution of the reference schematic is only suitable for the interface used in the reference schematic. If other interface functions need to be added to the reference schematic, the RK806S-5 LDO distribution needs to be re-evaluated, otherwise the added functions may exceed the maximum current provided by the LDO



Power Sequence



Power description

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC5V0_SYS_S5	RK806_BUCK1	6.5A	VDD_CPU_BIG_S0	Slot:2	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK2	5A	VDD_NPU_S0	Slot:2	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK3	5A	VDD_CPU_LIT_S0	Slot:2	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK4	5A	VCC_3V3_S3	Slot:5	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK5	3A	VDD_GPU_S0	Slot:2	ADJ FB=0.5V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK6	3A	VDDQ_DDR_S0	Slot:6	ADJ FB=0.5V	ON	0.61V-LP4/4x 0.51V-LP5	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK7	3A	VDD_LOGIC_S0 VDD_LOGIC_MEM_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK8	3A	VCC_1V8_S3	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK9	3A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	1.1V-LP4/4x 1.05V-LP5	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK10	3A	VDD_DDR_S0	Slot:2	0.85V DVFS	ON	0.85V DVFS	TBD	TBD
VCC_2V0_PLDO	RK806_PLDO1	0.5A	VCCA_1V8_S0	Slot:3	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO2	0.3A	VCCA1V8_PLDO2_S0	Slot:5	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	1.2V	TBD	TBD
VCC5V0_SYS_S5	RK806_PLDO4	0.5A	VCCA_3V3_S0	Slot:5	3.0V	ON	3.3V	TBD	TBD
	RK806_PLDO5	0.3A	VCCIO_SD_S0	Slot:5	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS_S5	RK806_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
	RK806_NLDO3	0.5A	VDDA0V75_HDMI_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	0.85V	TBD	TBD
	RK806_NLDO5	0.3A	VDDA_0V75_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_RESETn								
VCC5V0_SYS_S5	EXT BUCK	2A	VCC_2V0_PLDO_S3	Slot:1	2.1V	ON	2.0V	TBD	TBD
VCC5V0_SYS_S5	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	1.1V	TBD	TBD
VCC12V_DCIN	EXT BUCK	6A	VCC5V0_SYS_S5	Slot:0	5.0V	ON	5.0V	TBD	TBD
VCC12V_DCIN	EXT BUCK	3A	VCC5V0_DEVICE_S0	Slot:5A	5.2V	ON	5.2V	TBD	TBD
VCC_3V3_S3	SWITCH	2A	VCC_3V3_S0	Slot:5A	3.3V	ON	3.3V	TBD	TBD
VCC_1V8_S3	SWITCH	2A	VCCIO_1V8_S0	Slot:3A	1.8V	ON	1.8V	TBD	TBD

Note:

The power suffix S0, S3 or S5 means:
S5: Keep power on during power down
S3: Keep power on during sleeping
S0: Power off during sleeping

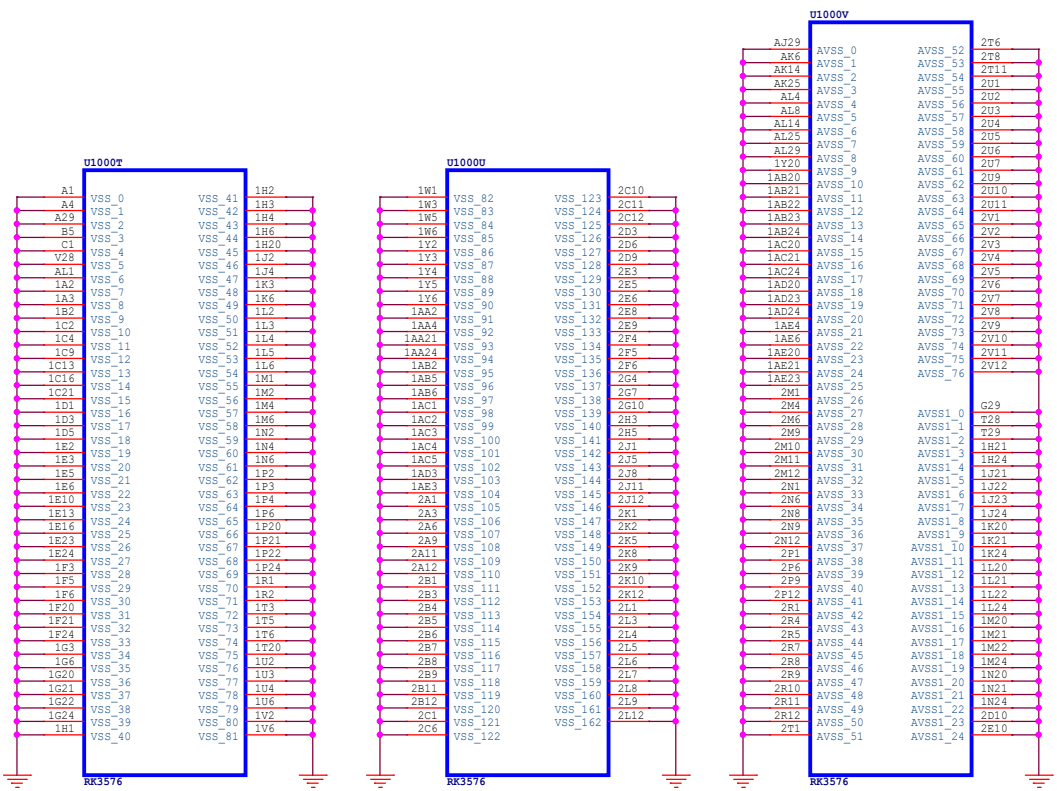
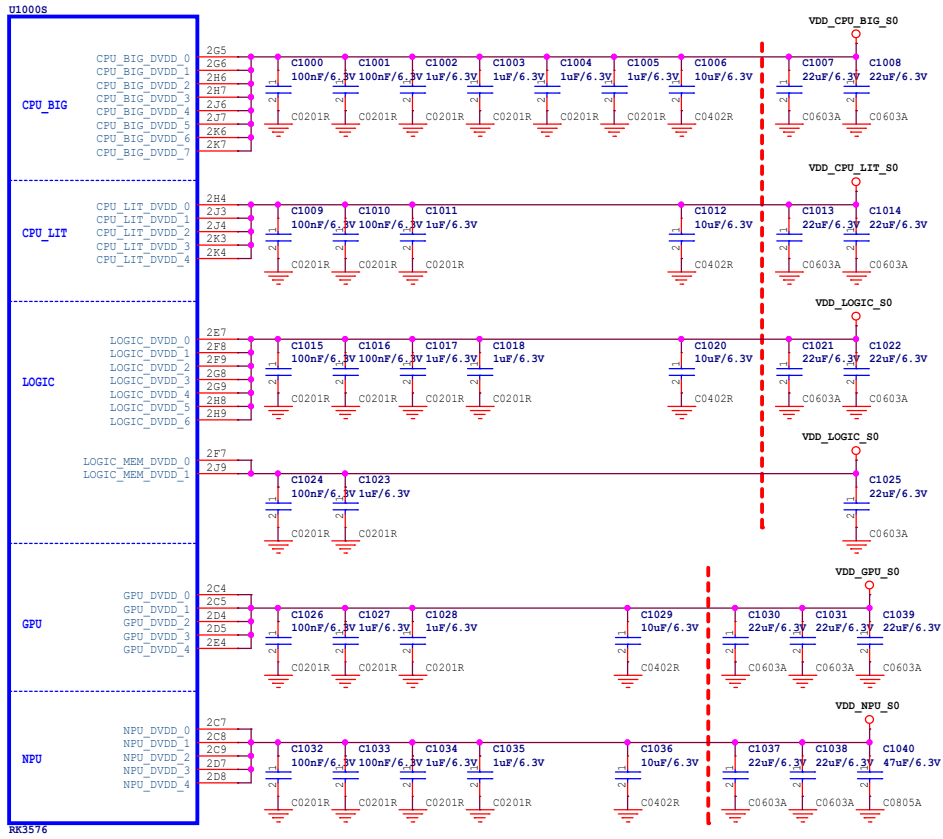
IO Power Domain Map

IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO0	Pin 2K11	1.8V Only	PMUIO0_VCC1V8	VCC_1V8	1.8V
PMUIO1	Pin 1U20	1.8V or 3.3V	PMUIO1_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO0	Pin 1J20	1.8V Only	VCCIO0_VCC1V8	VCC_1V8	1.8V
VCCIO1	Pin 2A8	1.8V or 3.3V	VCCIO1_VCC	VCC_1V8 VCC_3V3	1.8V/3.3V
VCCIO2	Pin 2A2	1.8V or 3.3V	VCCIO2_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO3	Pin 2B10	1.8V or 3.3V	VCCIO3_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO4	Pin 2A7	1.8V or 3.3V	VCCIO4_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO5	Pin 2A4/2A5	1.8V or 3.3V	VCCIO5_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO6	Pin 2N3	1.8V or 3.3V	VCCIO6_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO7	Pin 2M3	1.2V or 1.8V	VCCIO7_VCC	VCC_1V2 VCC_1V8	1.2V

IO Type	Operating Voltage
1.8V Only	VCCIO*_VCC1V8=1.8V
1.2V or 1.8V	VCCIO*_VCC=1.2V or 1.8V
1.8V or 3.3V	VCCIO*_VCC=1.8V or 3.3V

RK3576_S (Power)

RK3576_T/U/V (GND)



Note:
Caps of between dashed red lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

[illegible]

SAR



Item	Rup	Rdown	ADC Value	Boot Mode
Config1	NC	10K	0	USB (Maskrom mode)
Config2	10K	1.13K	416	FSP10->USB
Config3	10K	2.49K	816	FSP11_M0->EMMC->USB
Config4	10K	4.3K	1231	FSP11_M1->EMMC->USB
Config5	10K	6.8K	1658	FSP10->UFS->USB
Config6	10K	10K	2048	FSP11_M0->UFS->USB
Config7	10K	14.7K	2438	UFS->USB
Config8	10K	23.2K	2887	UFS->SDMMC0->USB
Config9	10K	40.2K	3279	RFU
Config10	10K	88.7K	3679	EMMC->SDMMC0->USB
Config11	10K	NC	4095	EMMC->USB

Boot From FSPI Or UFS/SD

D

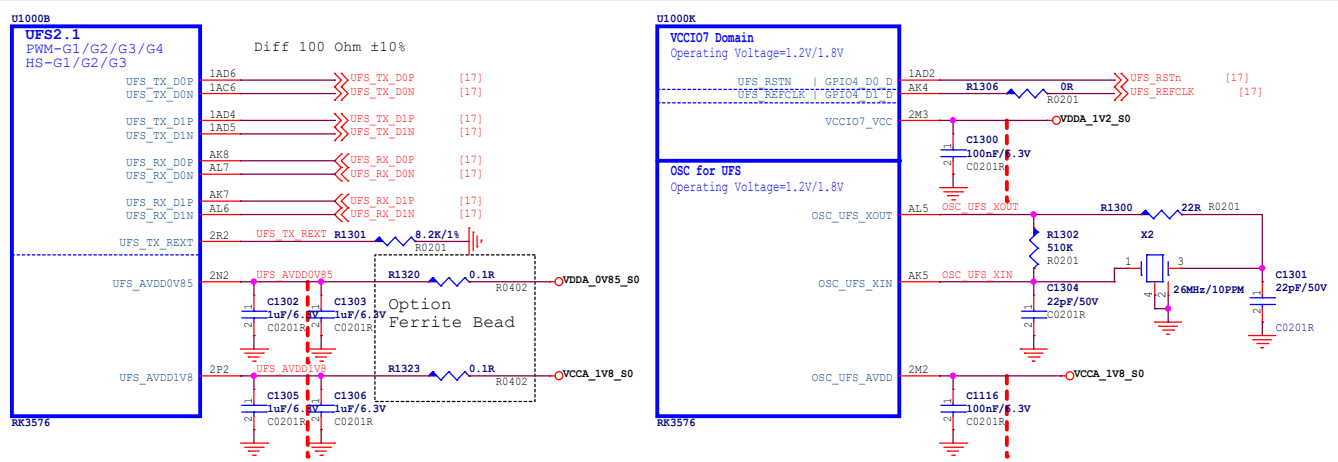


A



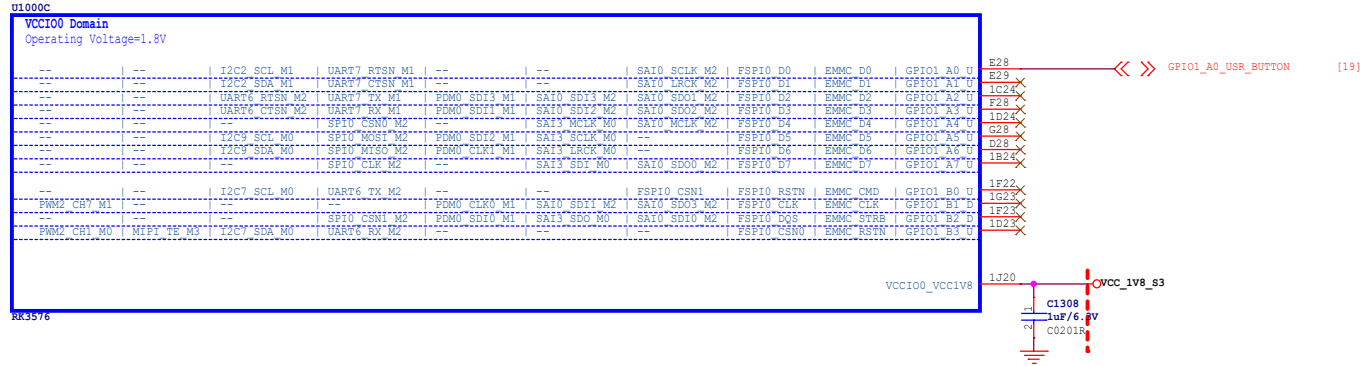
2

RK3576_B
(UFS2.1)



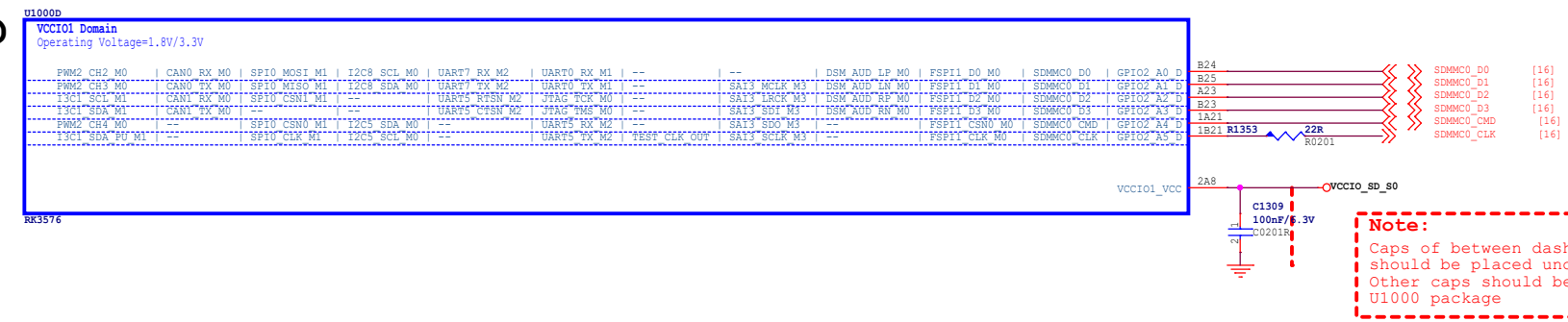
Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_C
(VCCIO0)



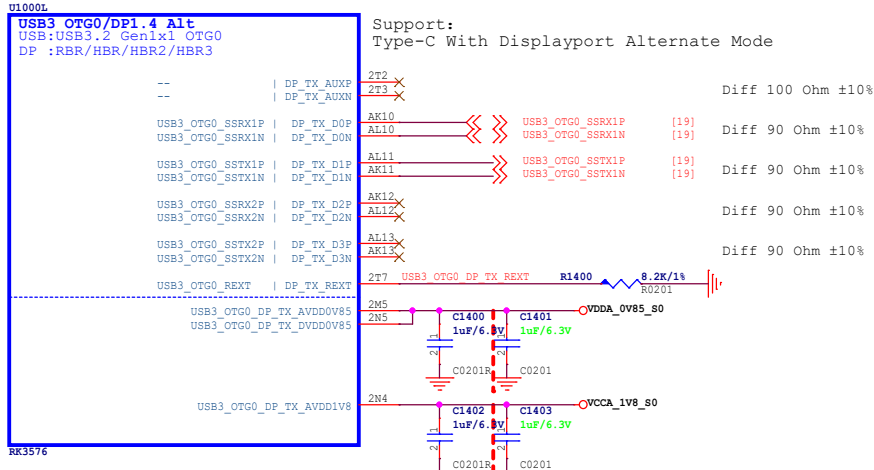
Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_D
(VCCIO1)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_L (USB3/DP)



Note:

Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_Q (HDMI/eDP)

Note:
HDMI 2.1 supports up to 4Kx2K@120Hz

U1000Q

HDMI TX/eDP TX Combo Phy
HDMI:V2.1 12Gbps
eDP :V1.3 5.4Gbps

Diff 100 Ohm ±10%

HDMI_TX_SBDP | EDP_TX_AUXP
HDMI_TX_SBDN | EDP_TX_AUXN

HDMI_TX_D0P | EDP_TX_D0P
HDMI_TX_D0N | EDP_TX_D0N

HDMI_TX_D1P | EDP_TX_D1P
HDMI_TX_D1N | EDP_TX_D1N

HDMI_TX_D2P | EDP_TX_D2P
HDMI_TX_D2N | EDP_TX_D2N

HDMI_TX_D3P | EDP_TX_D3P
HDMI_TX_D3N | EDP_TX_D3N

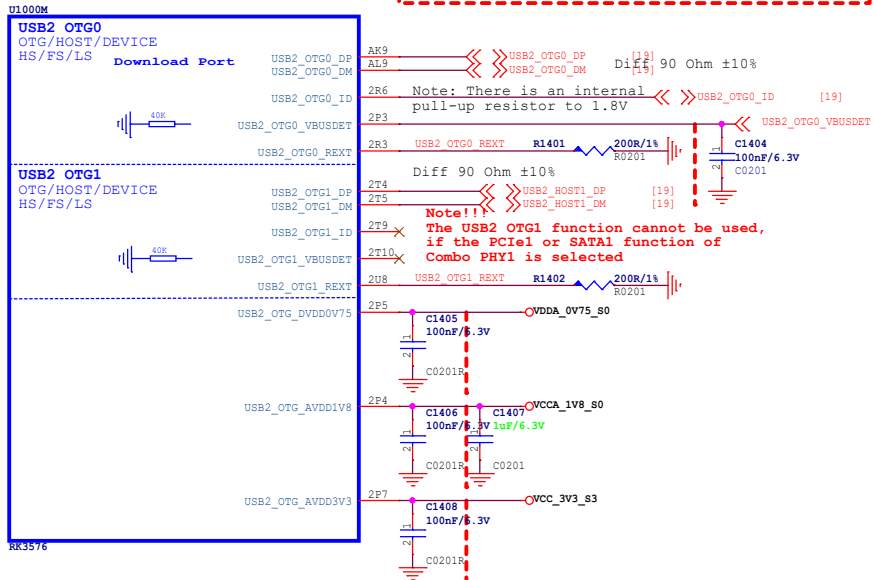
HDMI_TX_REXT | EDP_TX_REXT

2T12
2U12
1A24
AK27
AK28
AL28
AJ28
AK29
AL26
AK26
1AA20
2N11
2P11
2N10
2P10
C1600
C1601
C1602
C1603
C0201R
C0402R
VDDA_HDMI_S0
VCCA_1V8_S0
R1600
R0201
8.2K/1%

Note:

Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

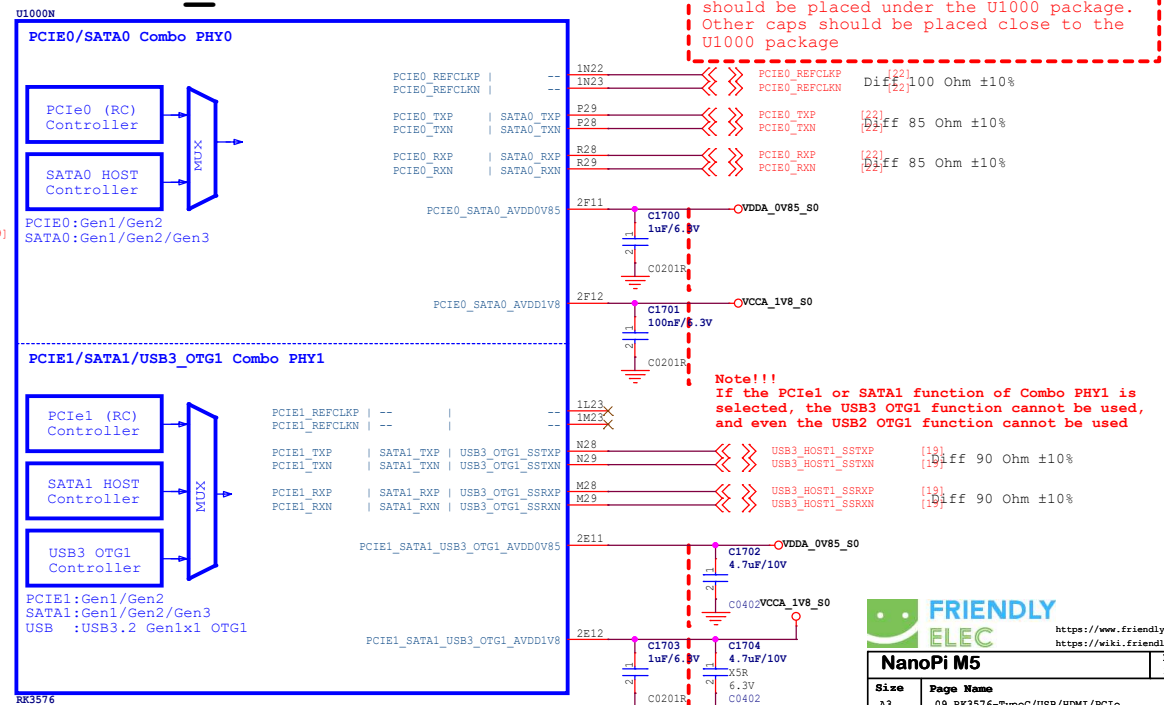
RK3576_M (USB2)



Note:

Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

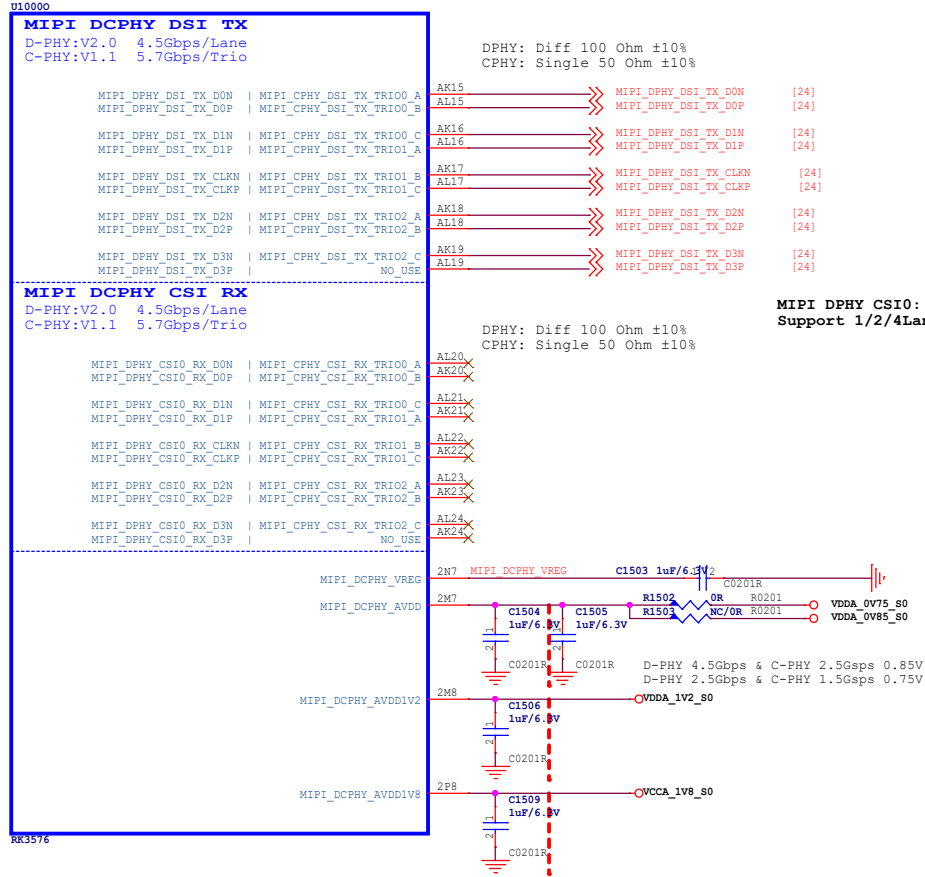
RK3576_N (PCIe/SATA/USB3)



Note:

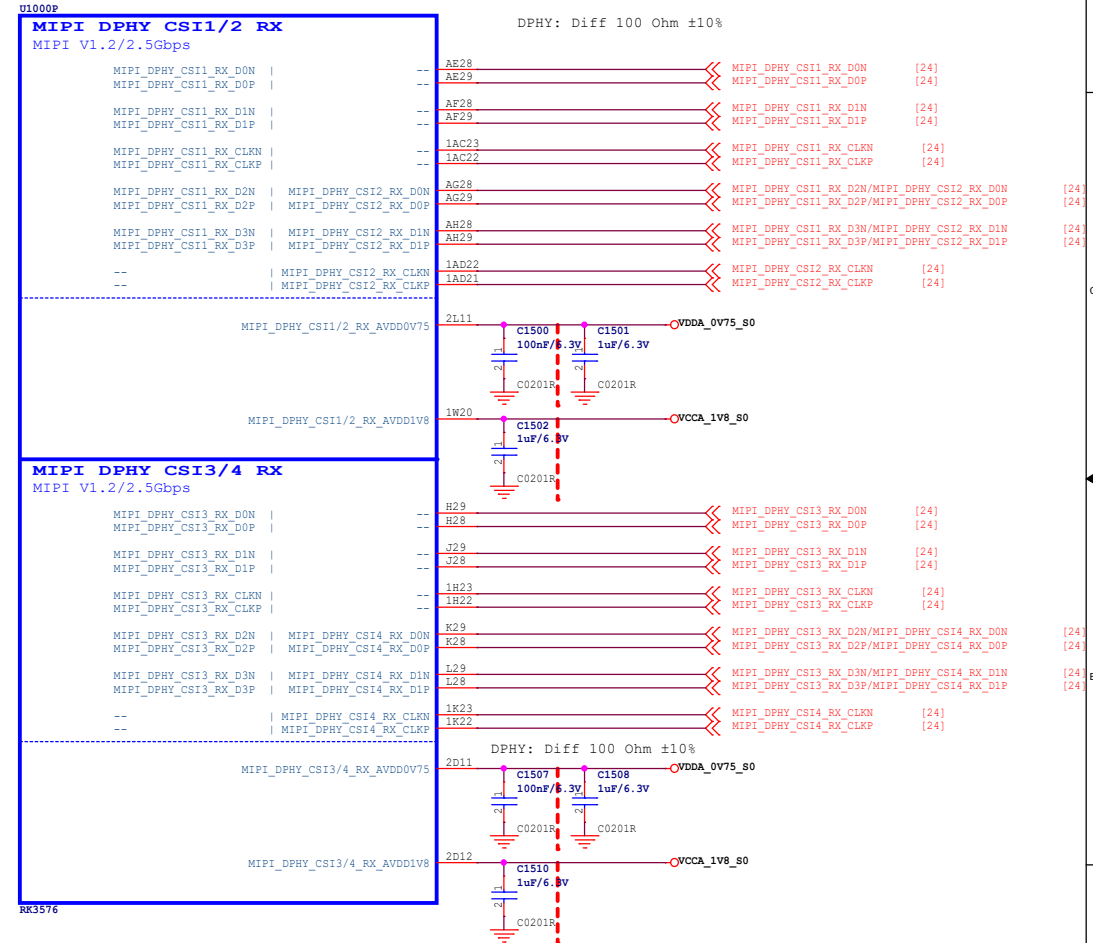
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_O (MIPI DCPHY)



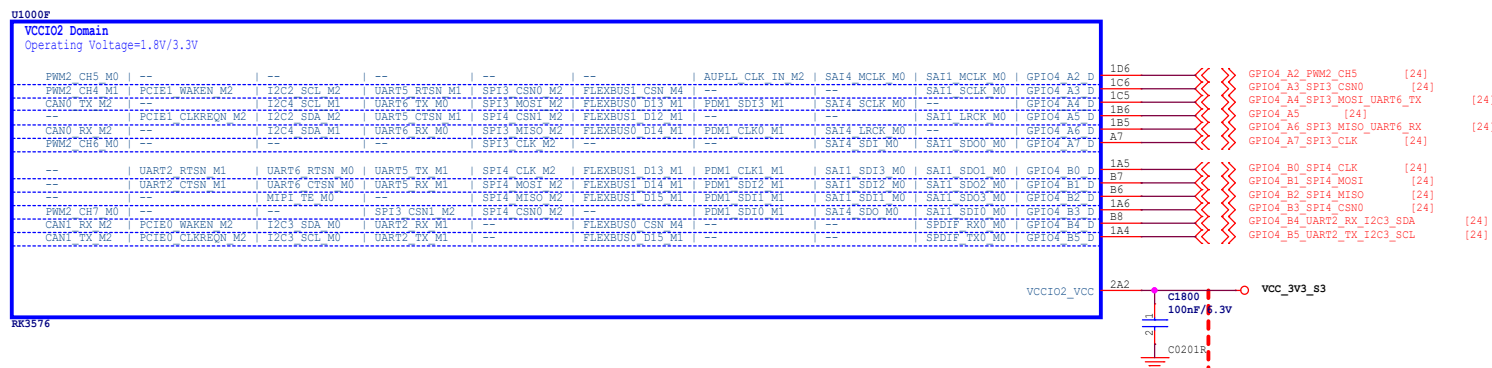
RK3576_P (MIPI DPHY CSI RX)

Support MIPI DPHY CSI1: 1/2/4Lane
Support MIPI DPHY CSI2: 1/2Lane
Support MIPI DPHY CSI3: 1/2/4Lane
Support MIPI DPHY CSI4: 1/2Lane
Support MIPI DPHY CSI1 2Lane + MIPI DPHY CSI2 2Lane



Support MIPI DPHY CSI3: 1/2/4Lane
Support MIPI DPHY CSI4: 1/2Lane
Support MIPI DPHY CSI1 2Lane + MIPI DPHY CSI2 2Lane
Support MIPI DPHY CSI3 2Lane + MIPI DPHY CSI4 2Lane

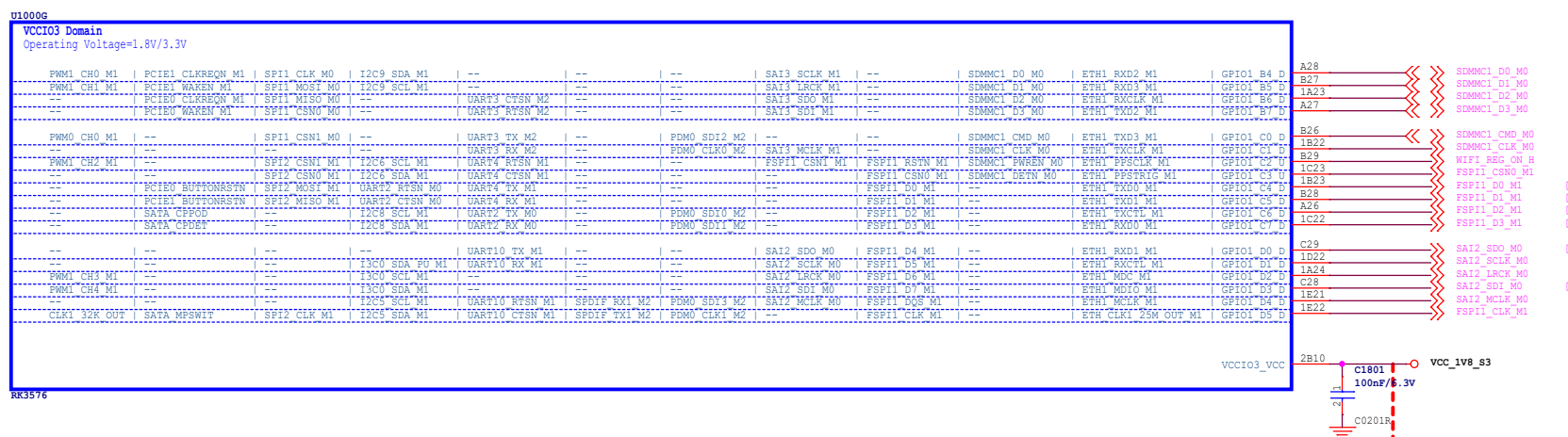
RK3576 F (VCCIO2)



! Note:

Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

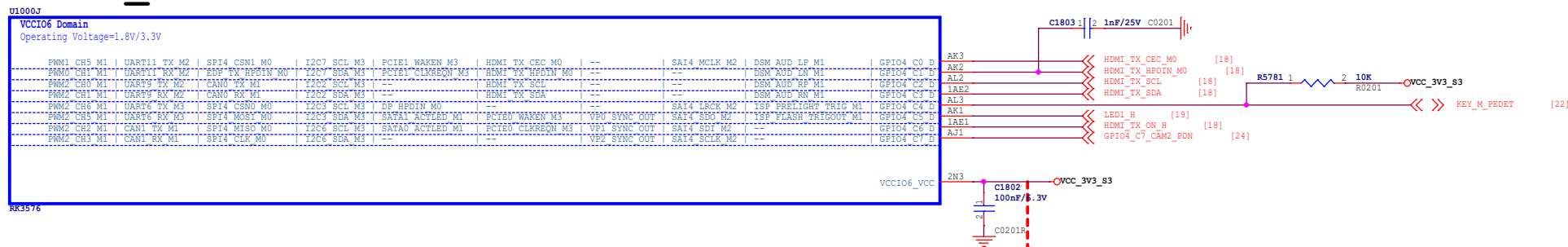
RK3576 G (VCCIO3)



Note:

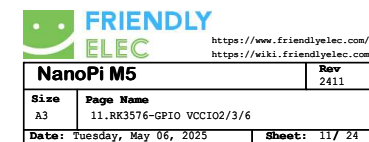
- Caps of between dashed red lines and U1000
- should be placed under the U1000 package.
- Other caps should be placed close to the
- U1000 package

RK3576 J (VCCIO6)

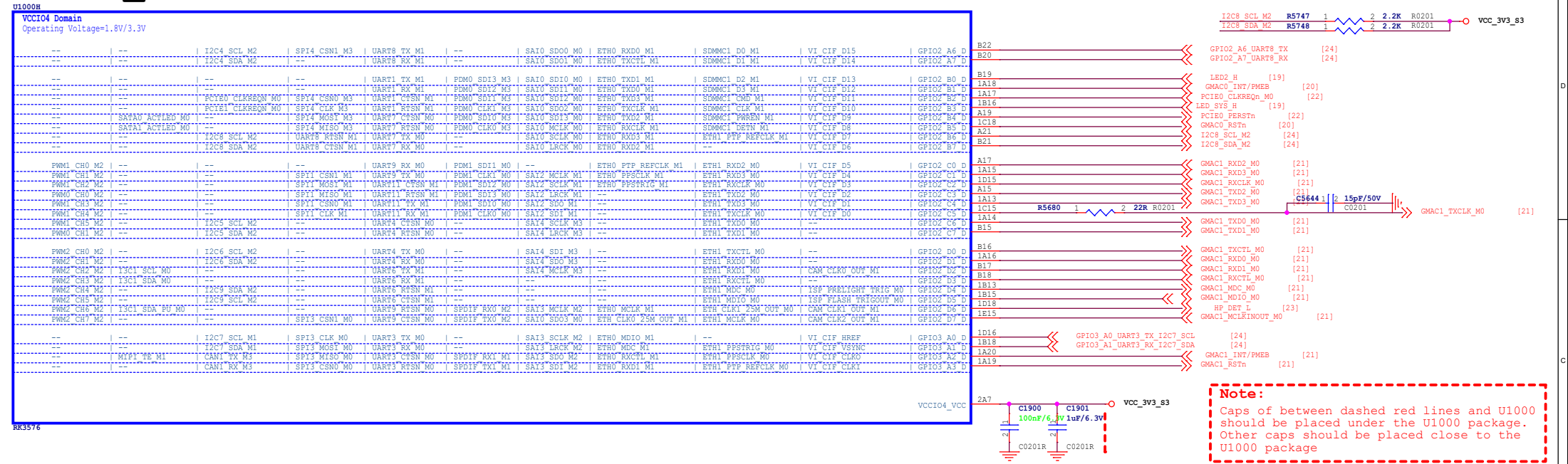


! Note:

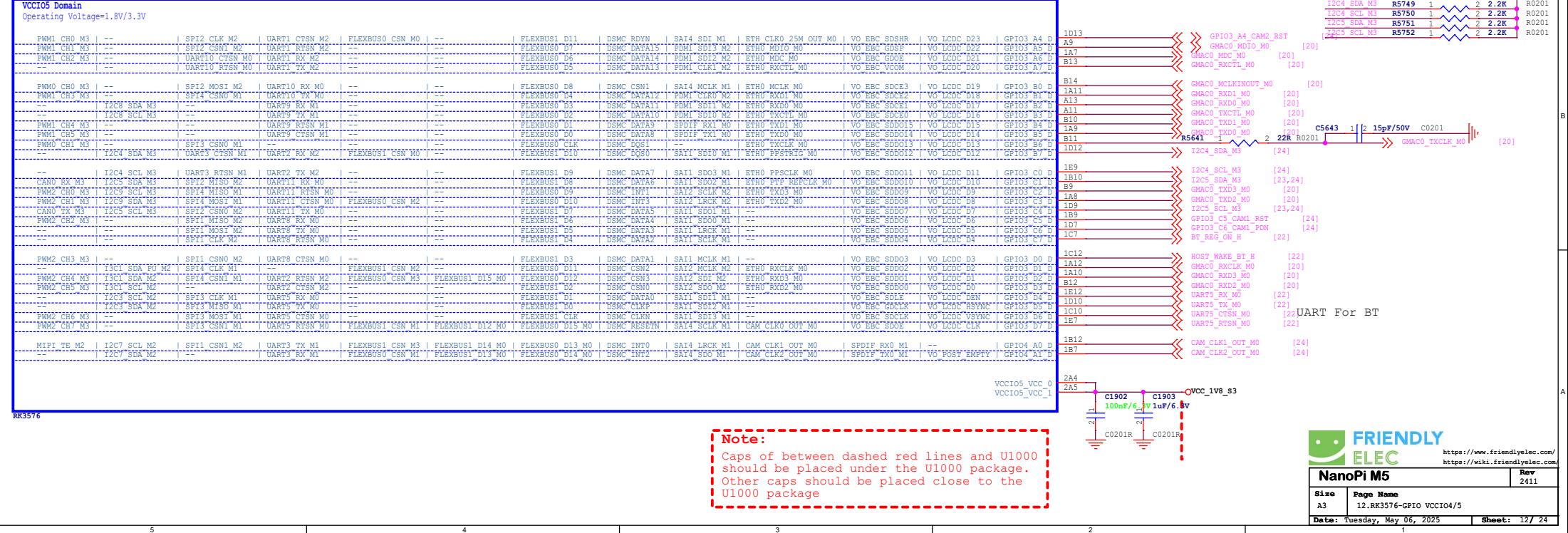
- Caps of between dashed red lines and U1000 should be placed under the U1000 package.
- Other caps should be placed close to the U1000 package



RK3576_H (VCCIO4)

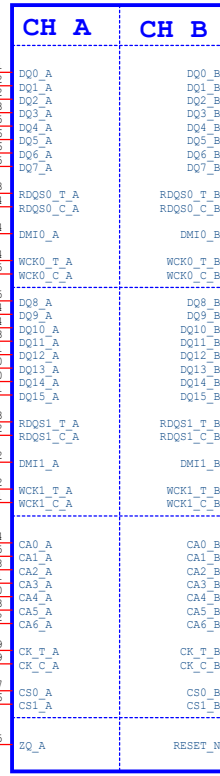


RK3576_I (VCCIO5)



LPDDR5

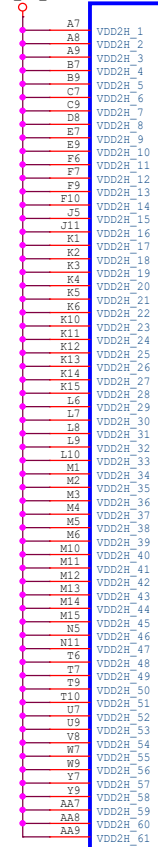
U3900A



LP5_B315

VDD2H_DDR_S3

U3900B



VDD2H
1.05V

VDD2L
0.9V

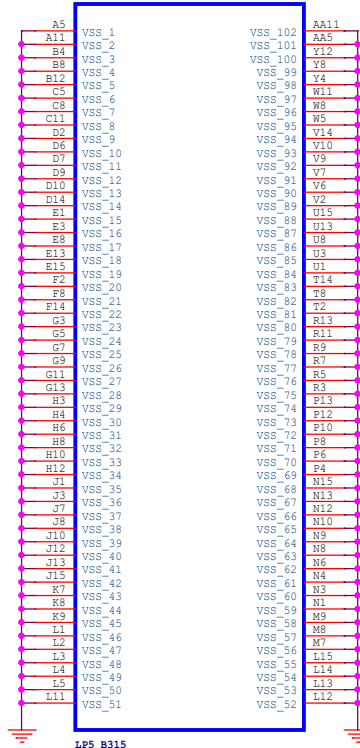
VDDQ
0.5V

VDD1
1.8V

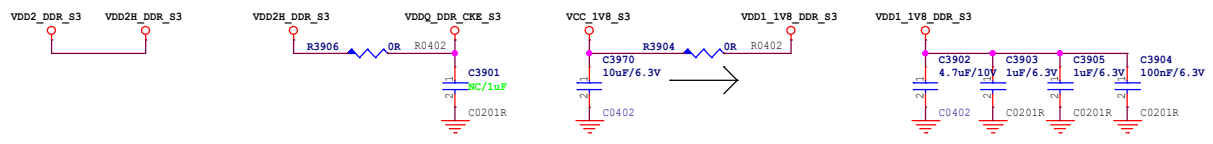
VDD2L_OV5_DDR_S3



U3900C



LP5_B315



Note:

Power Sequence: VDD1>VDD2H>VDD2L>VDDQ-200mV

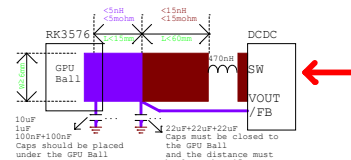
VDD1: 1.70V--1.95V, Power for core1.

VDD2H: 1.01V--1.12V, Power for core2 and Input Buffer.

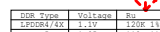
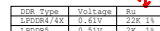
VDD2L: 0.87V--0.97V, Power for core2 and Input Buffer.

VDDQ: 0.47V--0.57V, Power for I/O Buffer.

I2C1_SDA_M0_RK806 <<>>
 I2C1_SCL_M0_RK806 <<>>
 FMIC_FWR_CTRL1 <<>>
 FMIC_FWR_CTRL2 <<>>
 [6] FMIC_INT_L <<>>
 [6,19] RESET_L <<>>
 FMIC_EXT_EN_OUT <<>>
 [15,19] FWRON_L <<>>



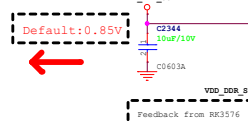
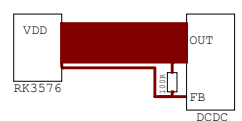
DO NOT DELETE IT!



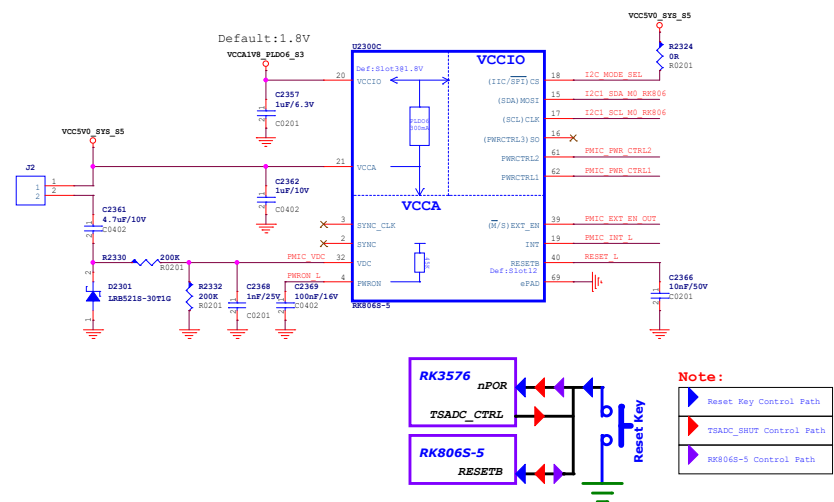
```

-----
Default:1.8V

```



```
Note:
I2C Mode:CS(pin18) connected to VCCA(pin21);
SPI Mode(Def):CS(pin18) floating or connected to GND
```



Note:

	Reset Key Control Path
	TSADC_SHUT Control Path
	RK806S-5 Control Path

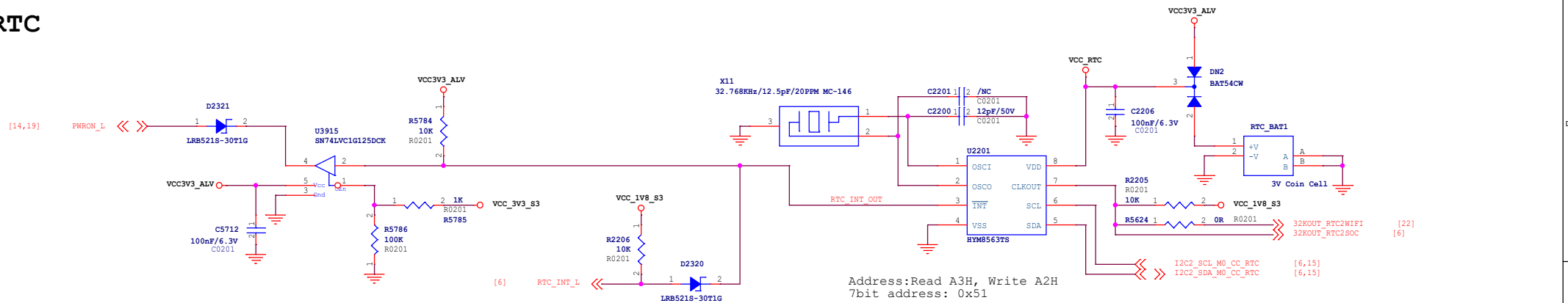
The schematic diagram illustrates the power supply architecture for the STM32F405RG microcontroller. It features two main internal regulators: the PLDO (Programmable Load Regulator) and the NLDO (Non-Linear Load Regulator). The PLDO is powered by the VCC2V0_P1D00_83 pin and provides multiple output rails: VCCA1V8_P1D00_80, VCCA1V8_P2D00_80, VDDA1V2_80, VCCA0V75_80, and VCCIO0_80_80. The NLDO is also powered by VCC2V0_P1D00_83 and provides output rails: VDD0V75_83, VDDA0V8_P1L1_80, VDDA0V8_P1M1_80, VDDA0V85_80, and VDDA0V75_80. Input capacitors (C2354, C2359, C2364, C2371) and output capacitors (C2353, C2355, C2356, C2358, C2360, C2363, C2365, C2367, C2370, C2372) are connected to the respective pins. The diagram also shows the internal capacitors C2354, C2359, C2364, and C2371, and the output capacitors C2353, C2355, C2356, C2358, C2360, C2363, C2365, C2367, C2370, and C2372. The internal regulators are labeled PLDO1 through PLDO5 and NLDO1 through NLDO5.

Note:

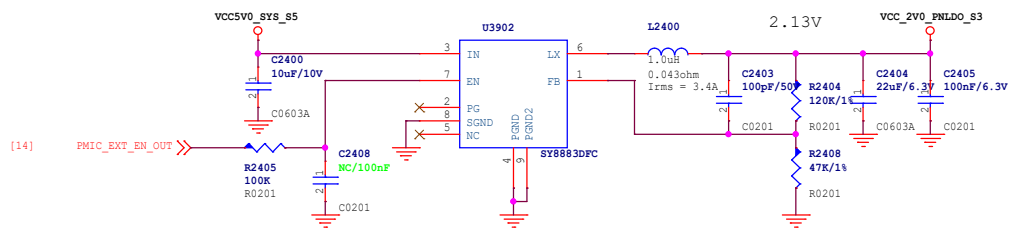
The RK806 LDO power distribution of the reference schematics is only suitable for the interface used in the reference schematics.

If other interface functions are to be added to the reference schematics, the RK806 LDO distribution needs to be re evaluated, otherwise the added

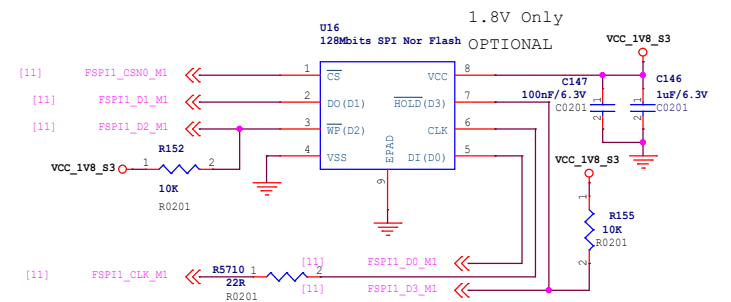
RTC



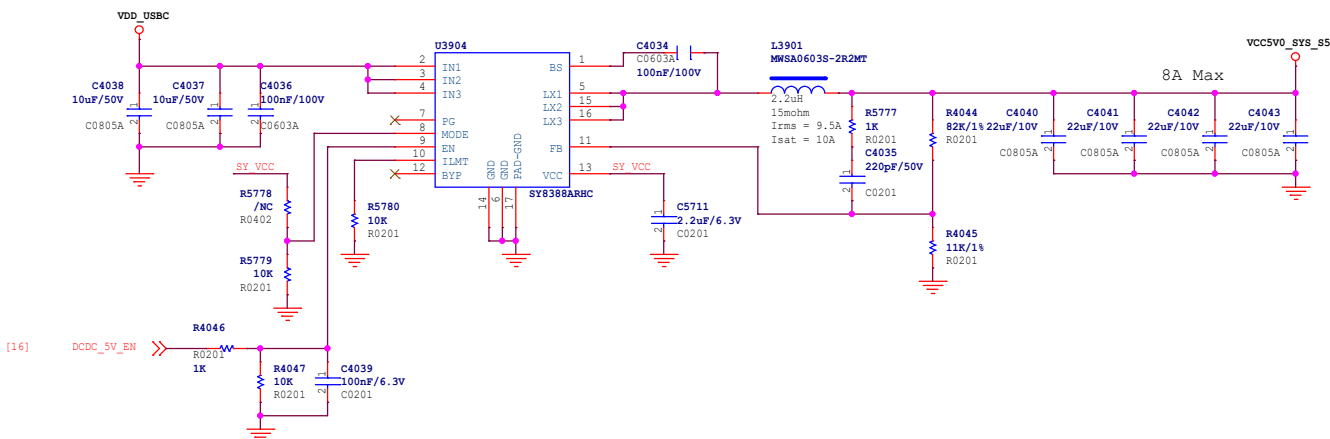
VCC_2V0_PLDO_NLDO_S3



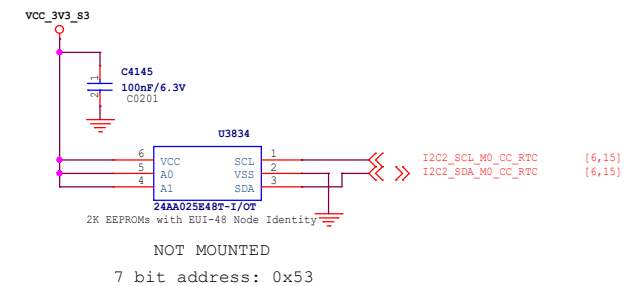
SPI Nor Flash



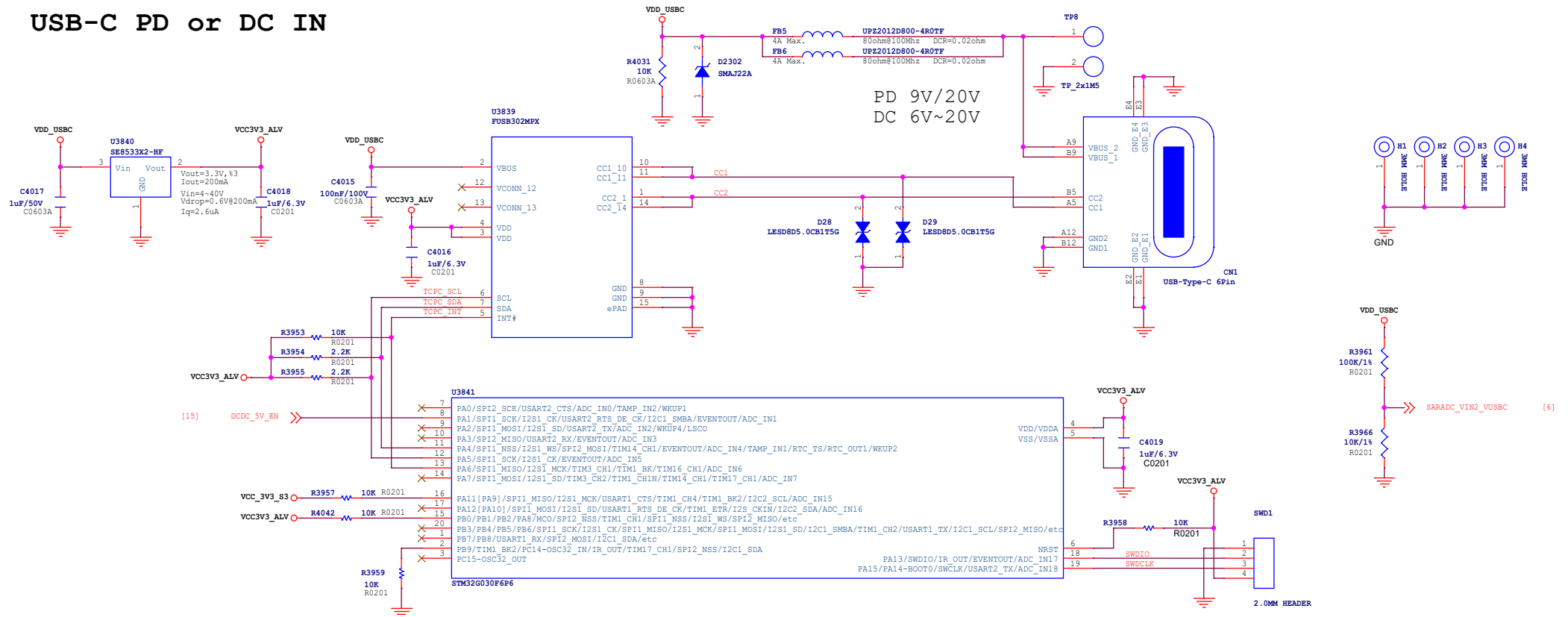
VCC5V0_SYS



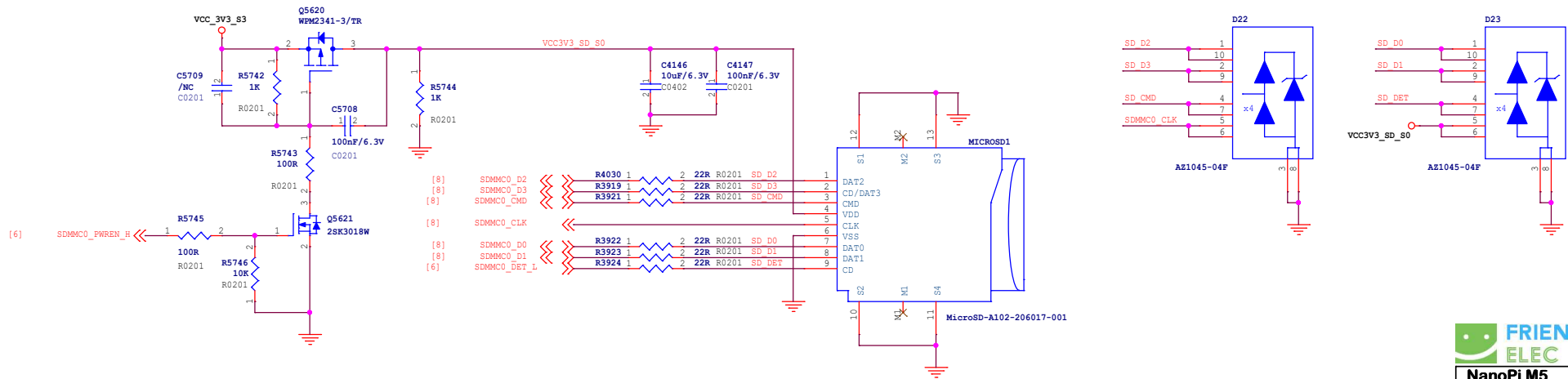
EUI-48 Node Identity



USB-C PD or DC IN

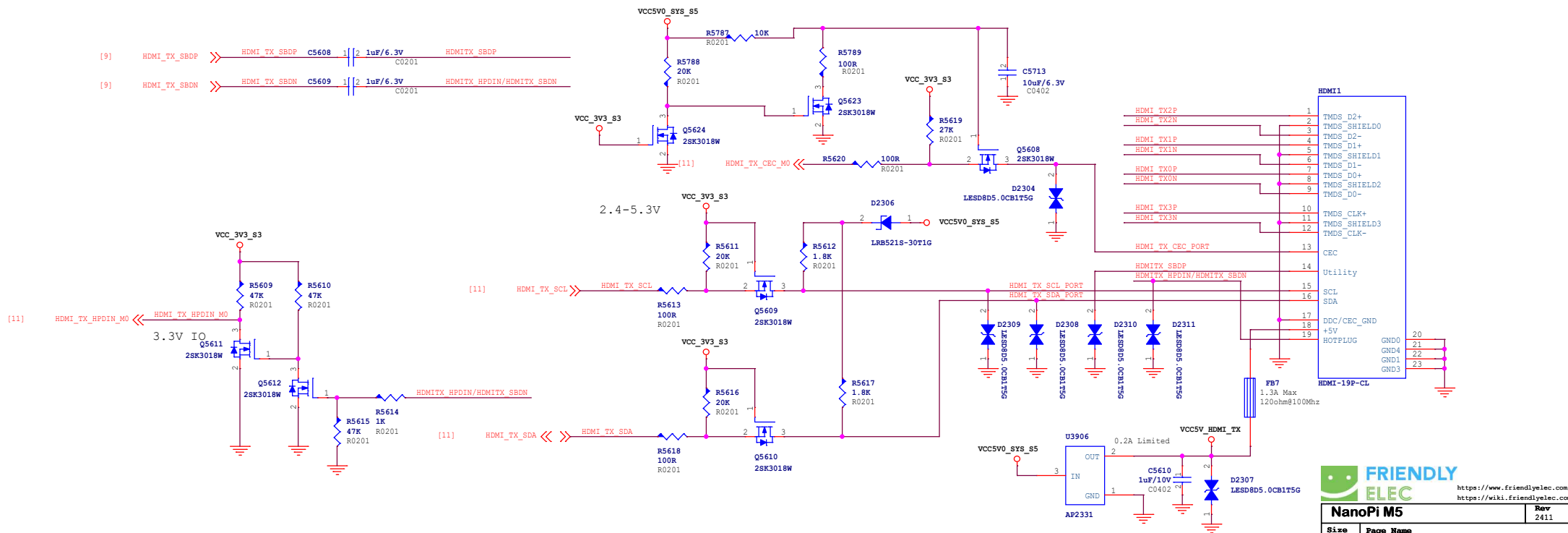
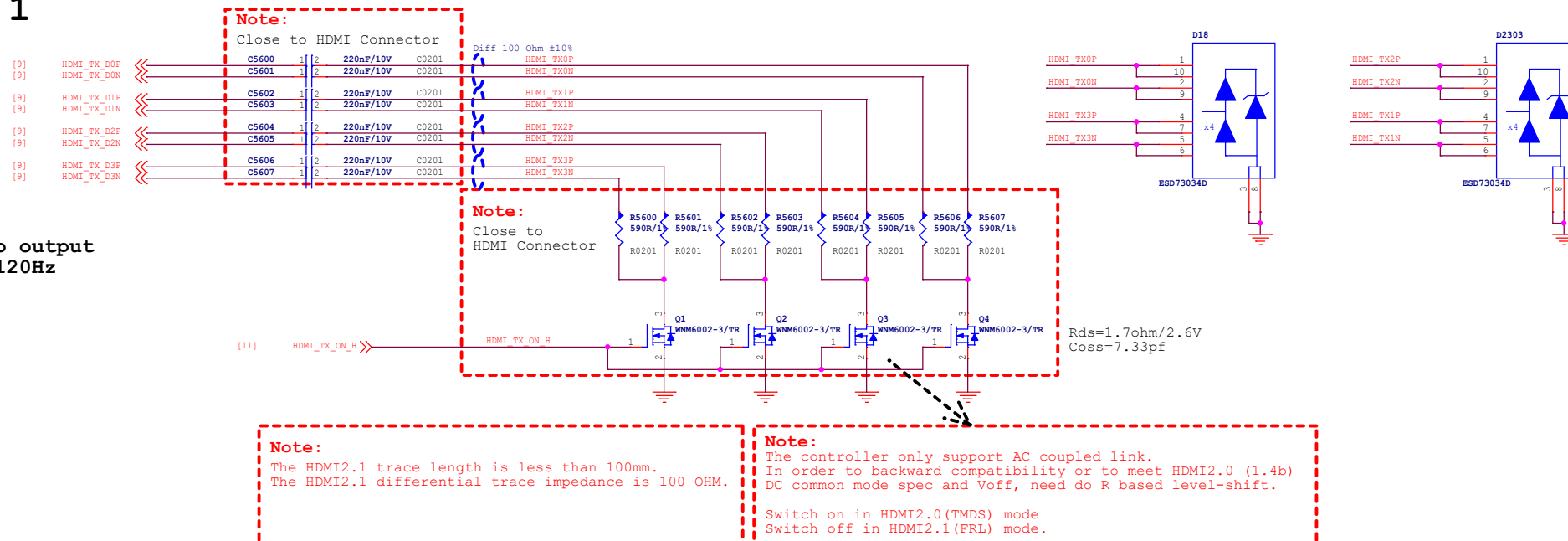


microSD

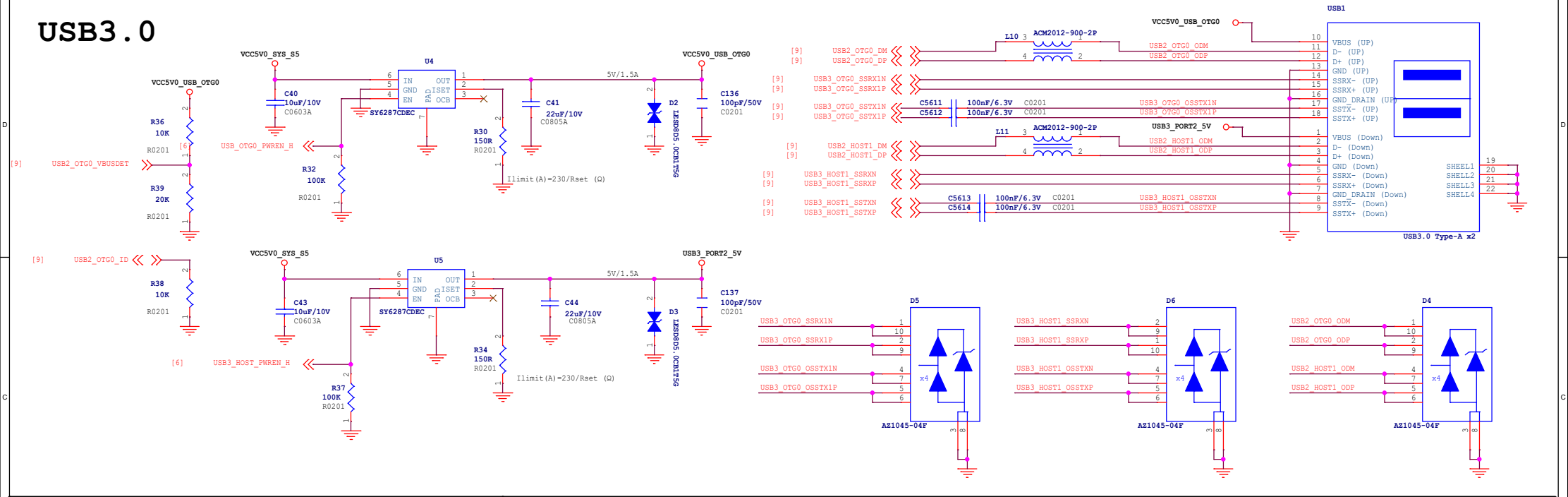


HDMI 2.1

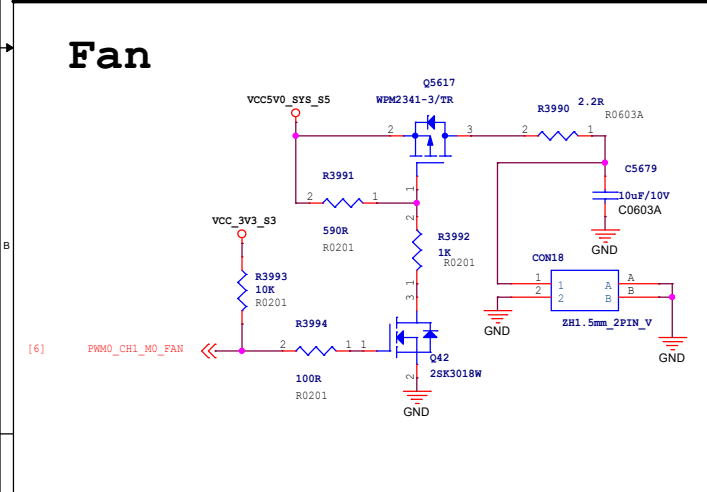
Support video output
up to 4Kx2K@120Hz



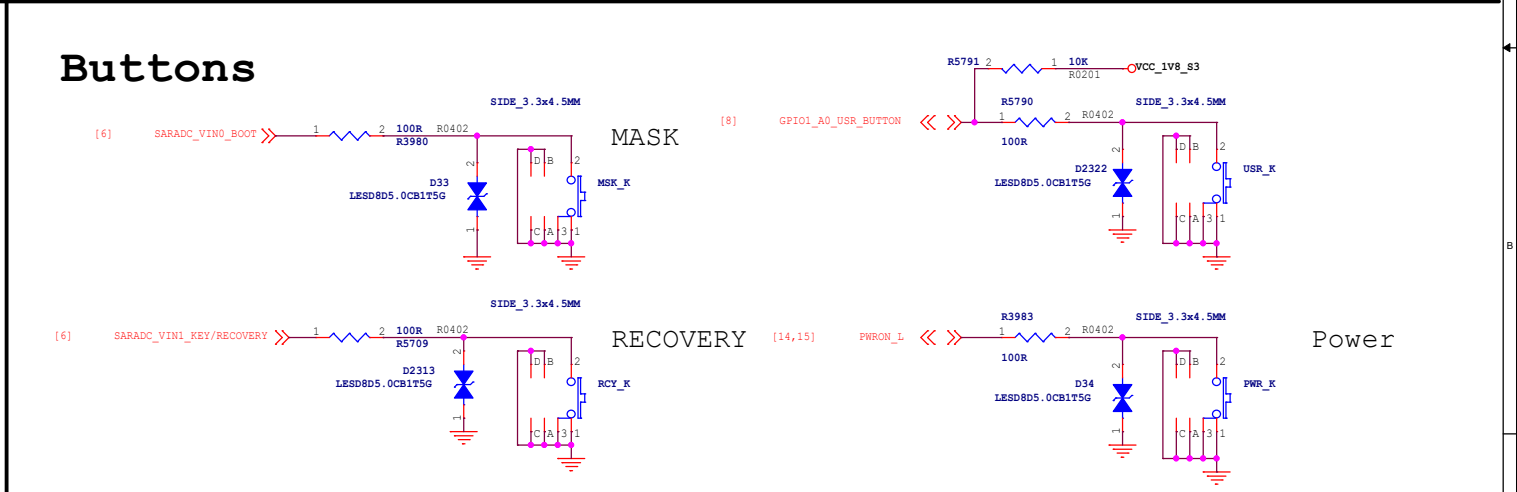
USB3.0



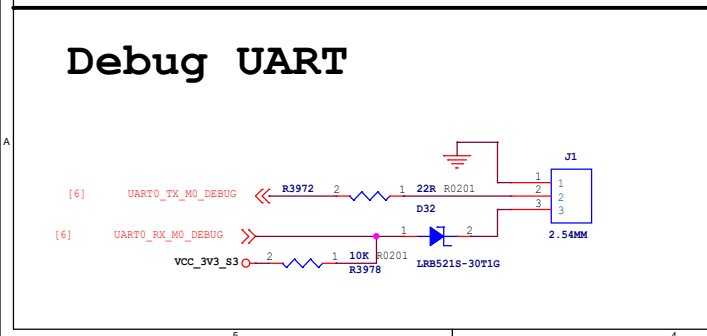
Fan



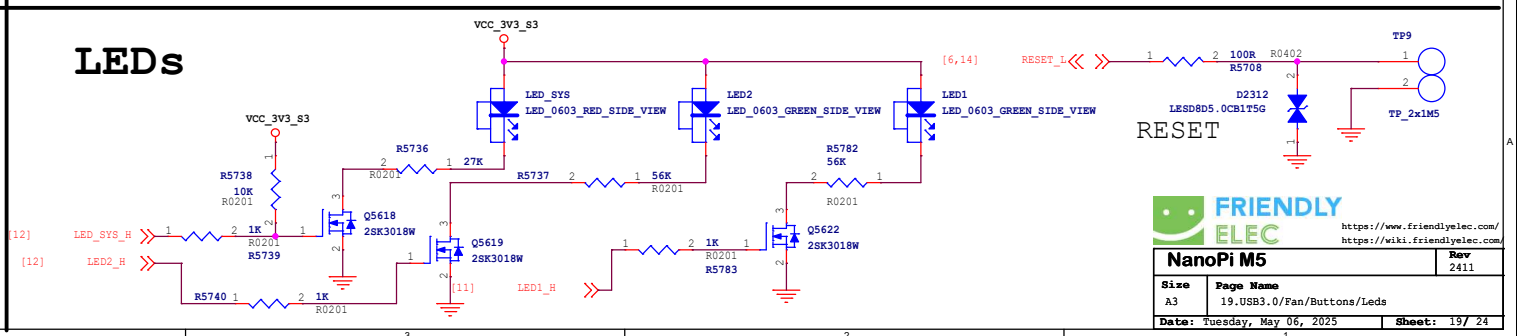
Buttons



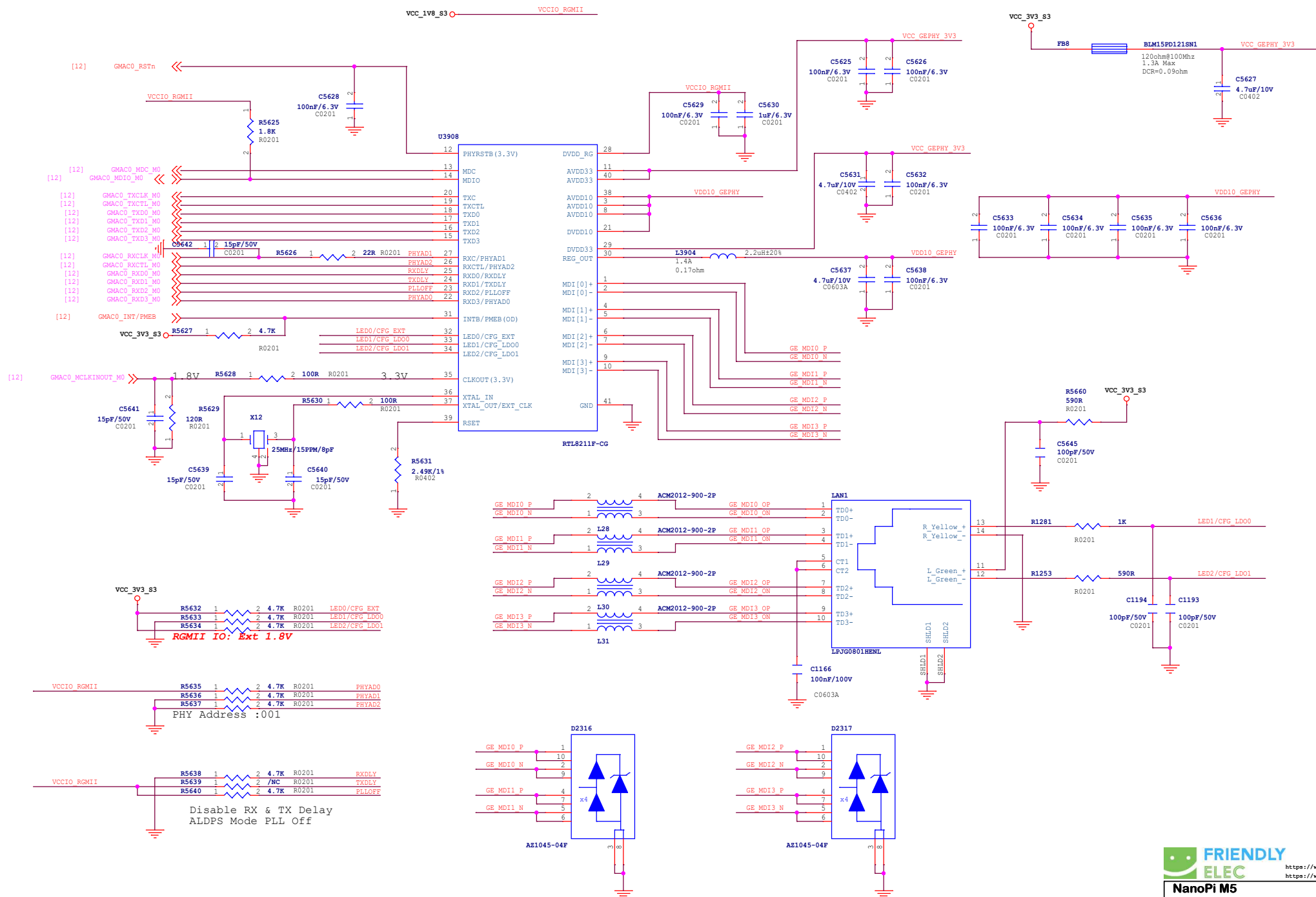
Debug UART



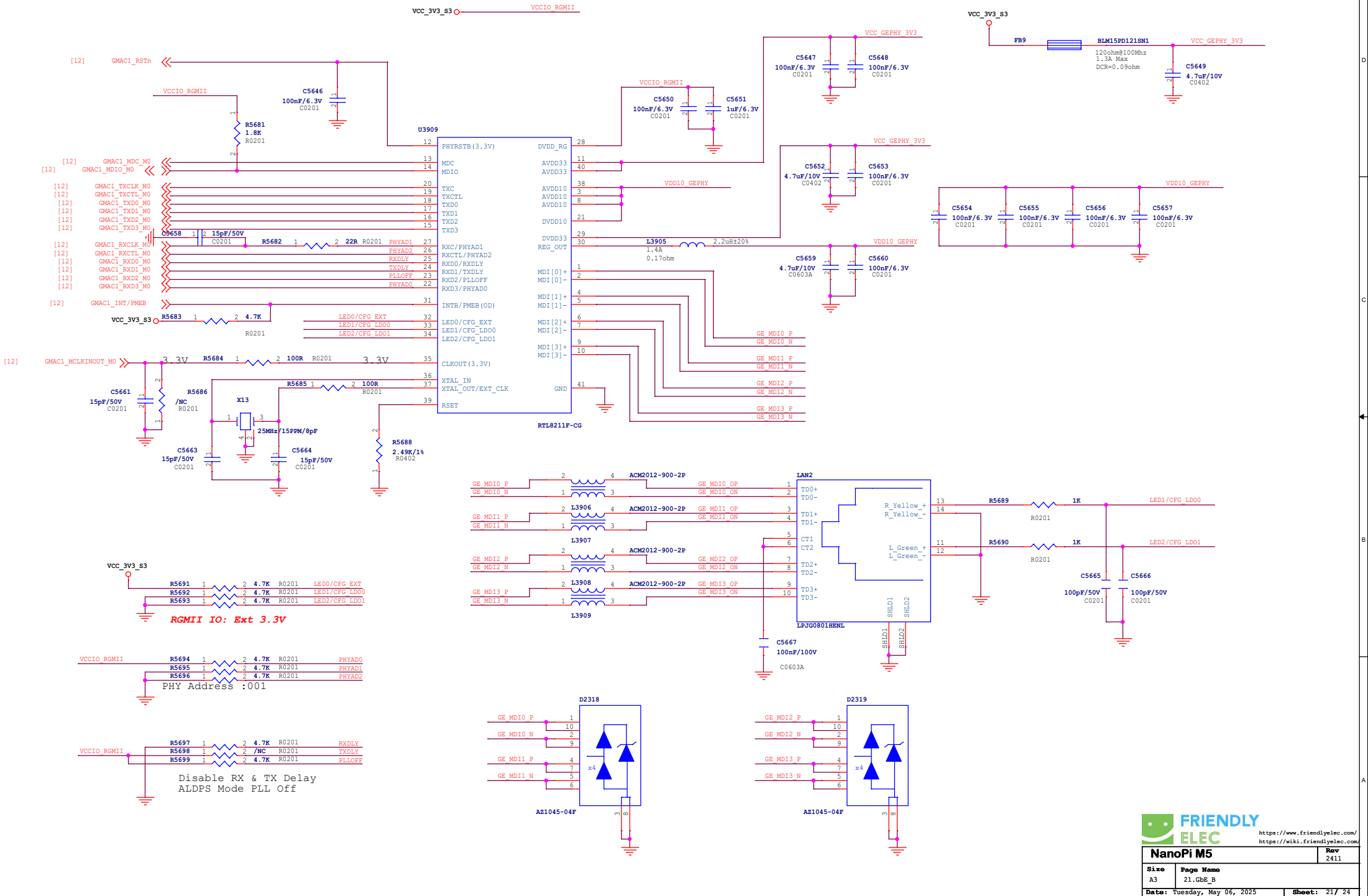
LEDs



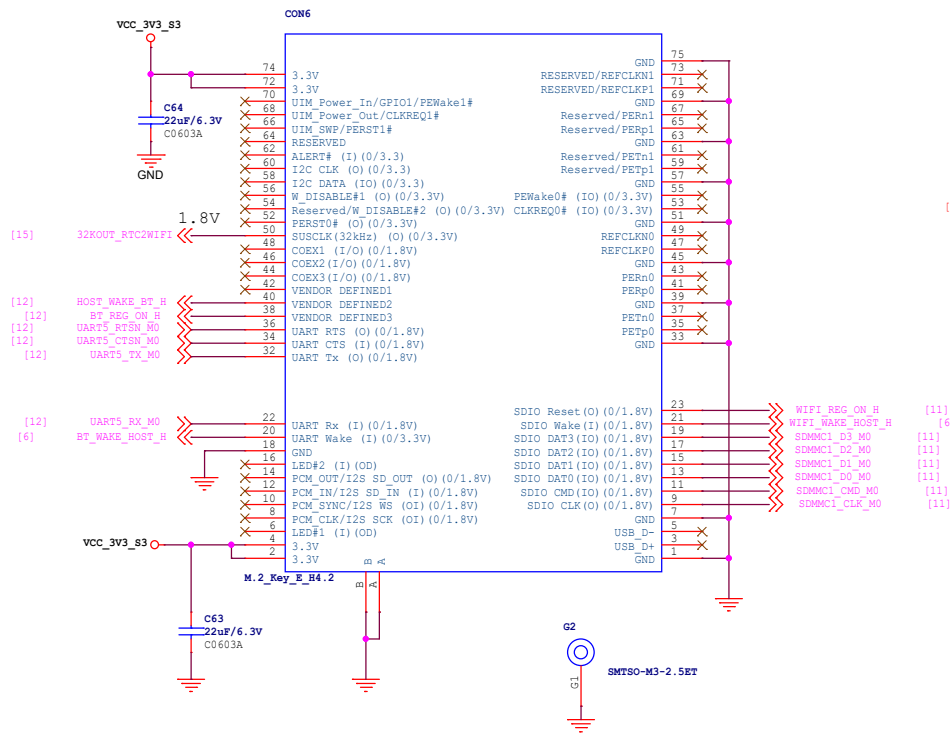
GbE A



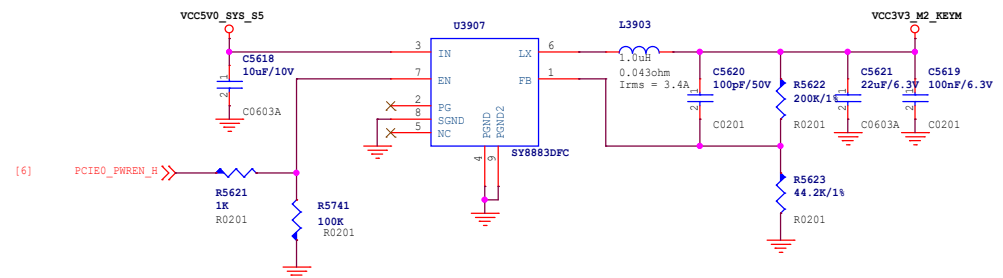
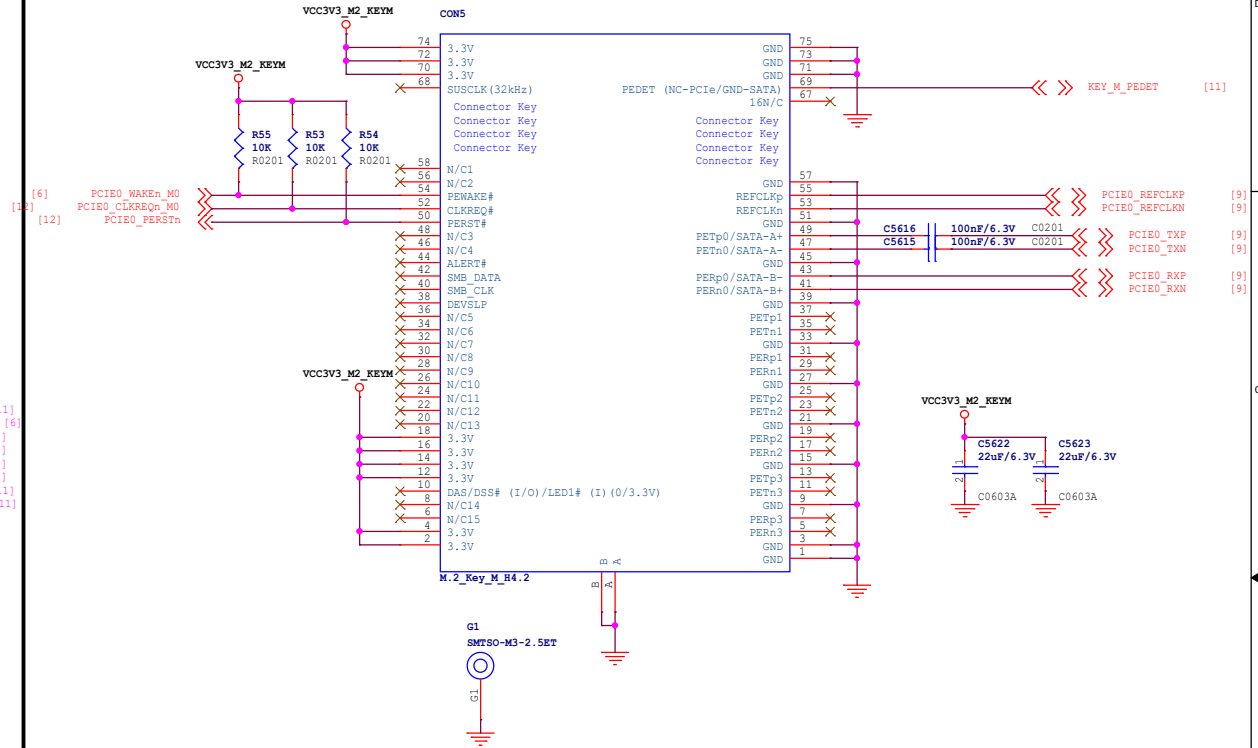
GbE B



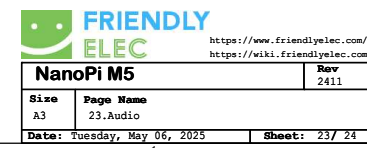
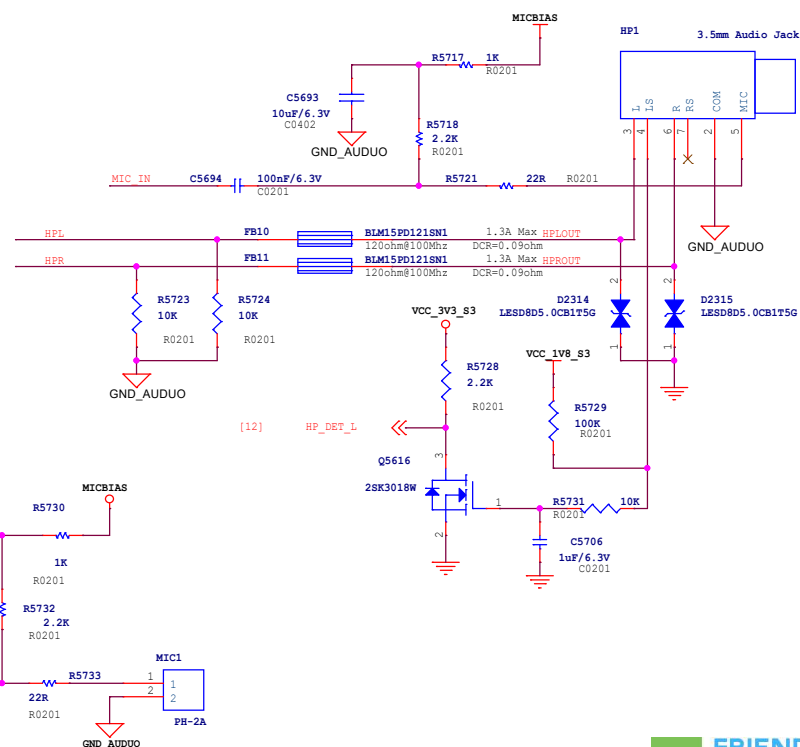
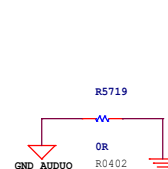
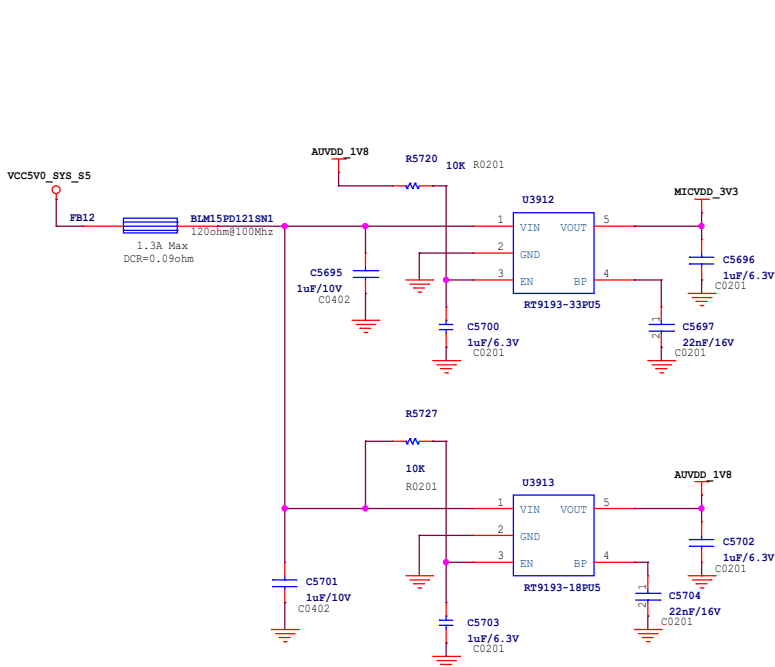
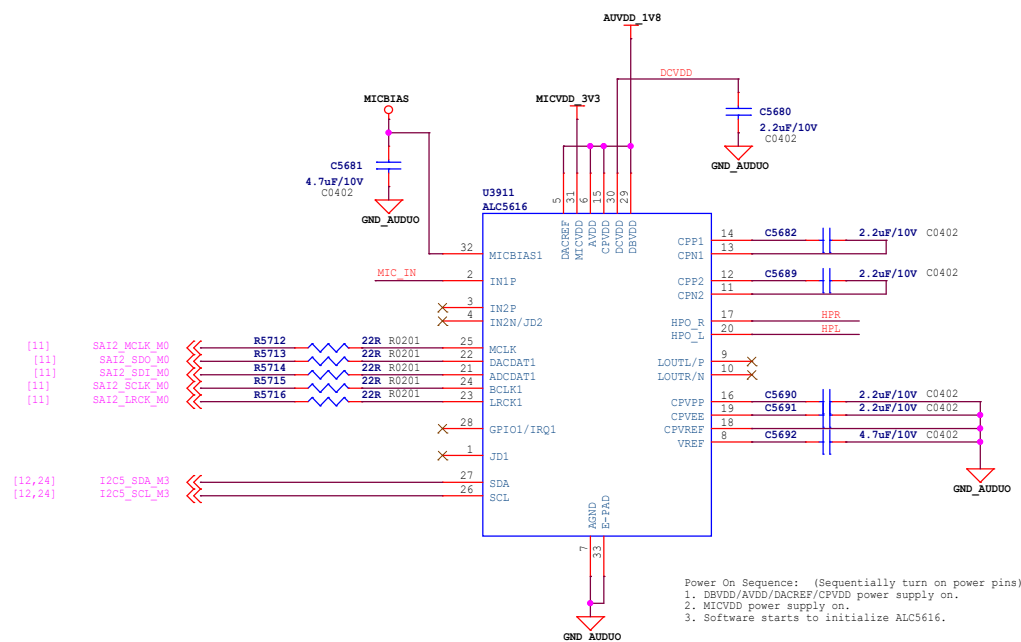
M.2 Key E 2230



M.2 Key M 2280



Audio

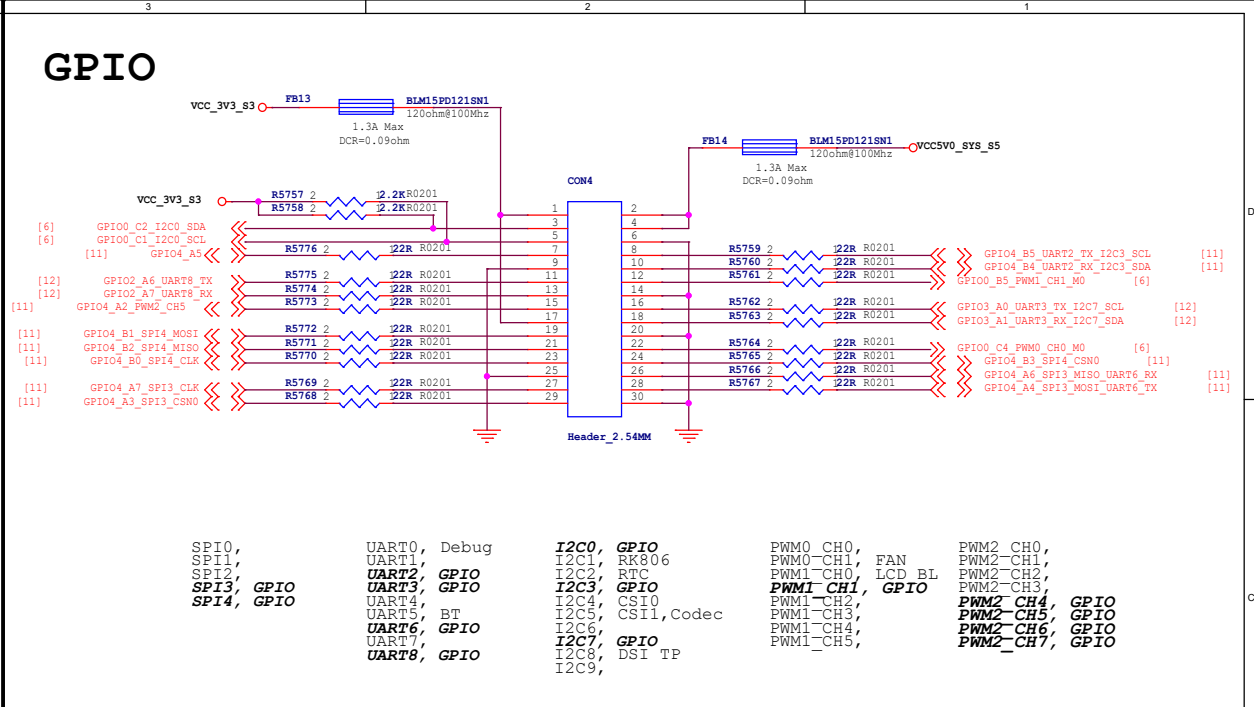


MIPI-CSIO

The diagram illustrates the MIPI-CSIO interface connections between a processor and a camera module (CON31). The processor side includes various pins and components:

- VCC5V0_SYS_S5**: Power supply pin connected to pin 1 of CON31.
- VCC1V8_S3**: Power supply pin connected to pin 6 of CON31.
- CAM_SYNC**: Synchronization signal pin connected to pin 7 of CON31.
- I2C4_SCL_M3** and **I2C4_SDA_M3**: I2C interface pins connected to pins 9 and 10 of CON31.
- GPIO3_C5_CAM1_RST** and **GPIO3_C6_CAM1_PDN**: Control pins connected to pins 11 and 12 of CON31.
- CAM_CLK1_OUT_M0**: Clock signal pin connected to pin 13 of CON31 via a resistor network (R5753 2, 22R R0201).
- MIPI_DPHY_CS11_RX_D3P/MIPI_DPHY_CS12_RX_DI1**: Data pins connected to pins 16 and 17 of CON31.
- MIPI_DPHY_CS11_RX_D3N/MIPI_DPHY_CS12_RX_DI0**: Data pins connected to pins 18 and 19 of CON31.
- MIPI_DPHY_CS11_RX_D2P/MIPI_DPHY_CS12_RX_D0P**: Data pins connected to pins 20 and 21 of CON31.
- MIPI_DPHY_CS11_RX_D2N/MIPI_DPHY_CS12_RX_D0N**: Data pins connected to pins 22 and 23 of CON31.
- MIPI_DPHY_CS11_RX_DI1** and **MIPI_DPHY_CS11_RX_DI0**: Data pins connected to pins 24 and 25 of CON31.
- MIPI_DPHY_CS11_RX_CLKP** and **MIPI_DPHY_CS11_RX_CLKN**: Clock signal pins connected to pins 26 and 27 of CON31.
- MIPI_DPHY_CS11_RX_D0P** and **MIPI_DPHY_CS11_RX_D0N**: Data pins connected to pins 28 and 29 of CON31.

The camera module (CON31) is a 30-pin connector with pins numbered 1 to 30. It includes two ground connections labeled A and B. The module is identified as **FPC 0.5MM_30PIN**.



MIPI-CSI1

[illegible]